

Python Operators – 7 Topics

Coding Programs with Output

1. Arithmetic Operators

Program:

```
a = 10
b = 3

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
print("Modulus =", a % b)
print("Exponent =", a ** b)
print("Floor Division =", a // b)
```

Output:

```
Addition = 13
Subtraction = 7
Multiplication = 30
Division = 3.3333333333333335
Modulus = 1
Exponent = 1000
Floor Division = 3
```

2. Assignment Operators

Program:

```
a = 10

a += 5
print("a += 5:", a)

a -= 3
print("a -= 3:", a)

a *= 2
print("a *= 2:", a)

a /= 4
print("a /= 4:", a)
```

Output:

```
a += 5: 15
a -= 3: 12
a *= 2: 24
a /= 4: 6.0
```

3. Comparison Operators

Program:

```
a = 10
b = 5

print("a == b:", a == b)
print("a != b:", a != b)
```

```
print("a > b:", a > b)
print("a < b:", a < b)
print("a >= b:", a >= b)
print("a <= b:", a <= b)
```

Output:

```
a == b: False
a != b: True
a > b: True
a < b: False
a >= b: True
a <= b: False
```

4. Membership Operators

Program:

```
numbers = [10, 20, 30, 40]

print("20 in numbers:", 20 in numbers)
print("50 in numbers:", 50 in numbers)
print("50 not in numbers:", 50 not in numbers)
```

Output:

```
20 in numbers: True
50 in numbers: False
50 not in numbers: True
```

5. Identity Operators

Program:

```
a = [10, 20]
b = a
c = [10, 20]

print("a is b:", a is b)
print("a is c:", a is c)
print("a is not c:", a is not c)
```

Output:

```
a is b: True
a is c: False
a is not c: True
```

6. Logical Operators

Program:

```
a = 10
b = 5

print("AND:", a > 5 and b < 10)
print("OR:", a > 15 or b < 10)
print("NOT:", not(a > b))
```

Output:

```
AND: True
OR: True
NOT: False
```

7. Bitwise Operators

Program:

```
a = 10
b = 5

print("AND:", a & b)
print("OR:", a | b)
print("XOR:", a ^ b)
print("Left Shift:", a << 1)
print("Right Shift:", a >> 1)
```

Output:

```
AND: 0
OR: 15
XOR: 15
Left Shift: 20
Right Shift: 5
```